Computer Networks – Lecture 1 & 2

Introduction, Network Types, Topologies, OSI & TCP/IP Models, Encapsulation

Introduction to Computer Networks

- A computer network is a system of interconnected devices
- Purpose:
-     Resource sharing
-     Communication
-     Data sharing & distributed processing
-     Internet and cloud access

Types of Networks

- PAN: Personal Area Network (Bluetooth, hotspot)
- BAN: Body Area Network (wearable sensors)
- LAN: Local Area Network (office, campus)
- MAN: Metropolitan Area Network (city-wide)
- WAN: Wide Area Network (Internet)

Network Topologies

- Bus: Simple, one cable, failure sensitive
- Star: Central hub/switch, easy to manage
- Ring: Closed loop, predictable
- Mesh: Full connectivity, reliable but costly
- Hybrid: Combination of two or more topologies

Layered Approach in Networking

Networking tasks divided into layers
Each layer has specific functions
Advantages:

- Modularity
- Standardization
- Interoperability
- Easier troubleshooting

OSI Reference Model (7 Layers)

7. Application – HTTP, FTP, DNS
6. Presentation – Encryption, compression
5. Session – Establish, maintain, terminate sessions
4. Transport – TCP/UDP, reliability
3. Network – IP, routing, addressing
2. Data Link – Framing, MAC, error correction
1. Physical – Transmission media, signals

TCP/IP Model (4 Layers)

- Application: HTTP, DNS, FTP, Email
- Transport: TCP, UDP
- Internet: IP, ICMP, ARP
- Network Access: Ethernet, Wi-Fi, PPP

OSI (7 layers) vs TCP/IP (4 layers)

Data Encapsulation (Sender)

- Application: User data created
- Transport: Adds TCP/UDP header → Segment/Datagram
- Network: Adds IP header → Packet
- Data Link: Adds Frame header/trailer → Frame
- Physical: Converts into bits → Transmission

Data Decapsulation (Receiver)

- Physical: Receives bits
- Data Link: Processes frame header, error check
- Network: Reads IP header → Packet
- Transport: Processes TCP/UDP header → Segment
- Application: Delivers data to user application

Data Units at Each Layer

- Application Layer → Message/User Data
- Transport Layer → Segments (TCP) / Datagrams (UDP)
- Network Layer → Packets
- Data Link Layer → Frames
- Physical Layer → Bits

Summary – Lecture 1 & 2

- Network Types: PAN, BAN, LAN, MAN, WAN
- Topologies: Bus, Star, Ring, Mesh, Hybrid
- Layered Approach: OSI & TCP/IP Models
- Data Encapsulation & Decapsulation
- Data Units: Message → Segment → Packet → Frame → Bits